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PKTA: Part-oriented Knowledge Transfer and Acquisition for Non-Exemplar Lifelong Person Re-Identification

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***Abstract*— Lifelong Person Re-identification (LReID) aims for a robust model on sequential data streams, adapting to dynamically growing tasks. Considering data privacy, this paper investigates the challenging Non-Exemplar LReID setting, which mitigates catastrophic forgetting without accessing previous-task exemplars. Existing methods typically adopt knowledge distillation to transfer old knowledge for anti-forgetting. However, they are coarse and lack sufficient exploration of pedestrian semantics, leading to the redundancy of useless old knowledge and thus impeding new task learning. To address these issues, we propose the Part-oriented Knowledge Transfer and Acquisition (PKTA) strategy, which delves into pedestrian parts information. Specifically, the Reliability-Oriented Processing (ROP) module refines more reliable pedestrian parts and feeds them into the Part Contribution Quantifier (PCQ) module, which quantifies each part's contribution to enable selective part-level knowledge transfer. Higher-contribution parts are favorable for learning new knowledge to distinguish newly added identities, while lower-contribution parts are inclined to transfer general old knowledge. Moreover, to suppress dynamic background noise during task transition, the Pedestrian-Aware Enhancement (PAE) module further guides the model to focus on pedestrian regions, ensuring high quality knowledge acquisition. Extensive experiments demonstrate the competitiveness of our method across seen and unseen domains.**

I. INTRODUCTION

Person Re-identification (ReID) [1], [2] is an image retrieval task that matches a query image with target images in a gallery. While remarkable performance has been attained in existing research, real-world deployment scenarios pose a critical constraint. Accessing all training data at once is impractical, since data usually arrives in incremental sequences. For example, surveillance systems typically provide sequential data streams. To tackle this constraint, Lifelong Person Re-identification (LReID) has been proposed to robustly identify the person over evolving data streams. A core challenge in LReID is catastrophic forgetting, which arises when models are updated with successive sequences of new data.

Most existing LReID methods rely on replaying old exemplars to retain old knowledge [22], [31]. However, due to data privacy constraints, re-accessing previous training data is impractical. This limitation gives rise to the more challenging Non-Exemplar LReID setting, which requires mitigating catastrophic forgetting without the access to old exemplars. Currently, most methods [18], [31], [28] in non-exemplar setting adopt knowledge distillation to transfer old

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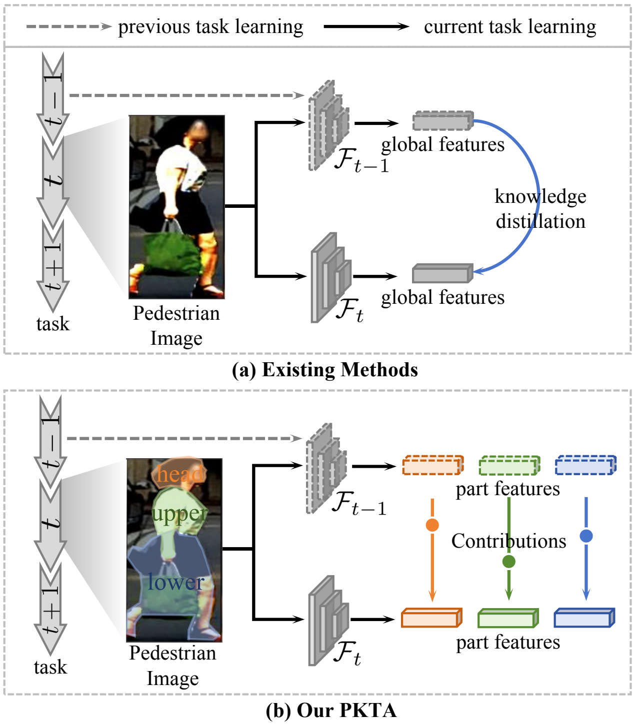

Fig. 1. Comparison between the existing method and our PKTA. (a) The existing method performs coarse knowledge transfer, which accumulates useless old knowledge and hinders new knowledge learning. (b) Our PKTA strategy performs selective part-level knowledge transfer to balance stability and plasticity.

knowledge for anti-forgetting. However, as shown in Figure 1 (a), these methods strictly align the global features of old and new models, neglecting pedestrian part details and indiscriminately transferring all old knowledge. This not only causes the redundancy of useless old knowledge, reducing model stability, but also occupies the learning space of new knowledge, impairing model plasticity.

Different pedestrian parts contribute distinctively to identity separability in the current task. We quantify the contribution of each pedestrian part across five benchmark datasets (Market-1501, CUHK-SYSU, DukeMTMC, MSMT17, and CUHK03) by evaluating the separability of features extracted solely from images that retain the corresponding body part, as presented in Figure 2. This key observation motivates the design of selective knowledge transfer strategies tailored to pedestrian parts with varying contributions, as illustrated in

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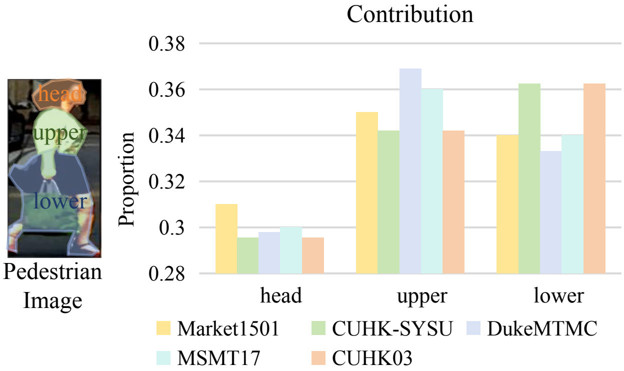


Fig. 2. Contribution of different pedestrian body parts to identity discriminability across five benchmark datasets.

Figure 1 (b). We can enable the model to preserve general knowledge of previously learned pedestrian identities while simultaneously avoiding interference with the acquisition of critical new identity knowledge, ultimately allowing us to strike an optimal balance between model stability and plasticity.

To this end, we propose a novel Part-oriented Knowledge Transfer and Acquisition (PKTA) strategy, which comprises three modules: Reliability-Oriented Processing (ROP) module, Part Contribution Quantifier (PCQ) module, and Pedestrian-Aware Enhancement (PAE) module. Specifically, ROP refines the initial pedestrian part semantics using the prior pixel distribution of each pedestrian part, yielding more reliable and semantically consistent part masks. These refined parts are then fed into PCQ, which efficiently measures the contribution of each part by evaluating the class separability of its part-guided features. The resulting part-level contributions then guide selective knowledge transfer. Higher-contribution parts underpin the discriminability of the new task, and the model should prioritize learning new knowledge linked to these parts. In contrast, lower-contribution parts serve to transfer general old knowledge, mitigating catastrophic forgetting. Furthermore, considering that dynamic background changes during task transitions introduce noise that interferes with pedestrian information exploration, PAE is introduced to leverage additional constraints derived from body features to strengthen the knowledge acquisition of pedestrian regions.

In summary, our contributions are as follows.

- We innovatively integrate structured pedestrian part information into selective part-level knowledge transfer, achieving the stability-plasticity trade-off.
- A novel pedestrian-aware enhancement module is designed to strengthen the acquisition of pedestrian region knowledge.
- Extensive experiments demonstrate the competitiveness of our proposed strategy. PKTA yields average performance gains by 1.7%/1.7% (mAP/R@1) on seen domains and 0.5%/0.7% (mAP/R@1) on unseen domains across two training orders.

II. Related Work

A. Person Re-identification

Person Re-identification (ReID) aims to address the problem of pedestrian identity matching across non-overlapping camera views [2]. In static data scenarios, some research has continually advanced ReID techniques. However, acquiring all training data at once is impractical in real-world scenarios, leading to poor generalization of ReID methods to unseen data and significant performance degradation [18], [32]. To address this challenge, Lifelong Person Re-identification (LReID) has been proposed to learn robust models from sequential data. Such methods are capable of both preserving previously learned pedestrian identities and adapting to novel identities.

B. Lifelong Person Re-identification

Lifelong Person Re-identification (LReID) aims to maintain robust identification capabilities across both previous and new pedestrians when learning from sequential data streams. Some works replay exemplars from prior tasks to preserve old knowledge [25]. KRC [31] uses a bi-directional strategy with both previous exemplars and new data to facilitate knowledge consolidation. Ge *et al.* [5] maintains knowledge consistency by modeling old tasks' distribution using exemplars. CLUDA-ReID [9] leverages exemplars to balance model adaptation and anti-forgetting. Despite their efficacy, these exemplar-based methods face challenges regarding data privacy and storage overhead.

Considering these limitations, this paper focuses on Non-Exemplar LReID, which achieves anti-forgetting without accessing previous exemplars. Current non-exemplar methods typically adopt knowledge distillation to transfer old knowledge to new models, *i.e.*, aligning the outputs of the new model with those of the old model. Some works [22], [4] perform patch-level distillation. Yet they require additional modules for patch sampling and perform distillation only on the selected patches, resulting in semantic absence. P^2-ARD [30] controls intra-class relation distillation using identity compactness. DKP[4] explores relation distillation between old prototypes and new features. And Huang *et al.* [8] extracts more discriminative regional features for knowledge distillation. However, the above methods lack sufficient exploration of pedestrians and transfer all old knowledge without selection. This leads to old knowledge, which occupies the new knowledge learning space and thereby hinders the new task learning. Conversely, our method fully considers various granular body parts and conducts selective knowledge transfer based on their contributions to identity separability.

III. Method

Figure 3 illustrates the pipeline of PKTA. **ROP** provides reliable pedestrian part semantics for **PCQ** to measure the part-level contributions and enable selective knowledge transfer. **PAE** prioritizes exploring pedestrian regions for high quality knowledge acquisition.

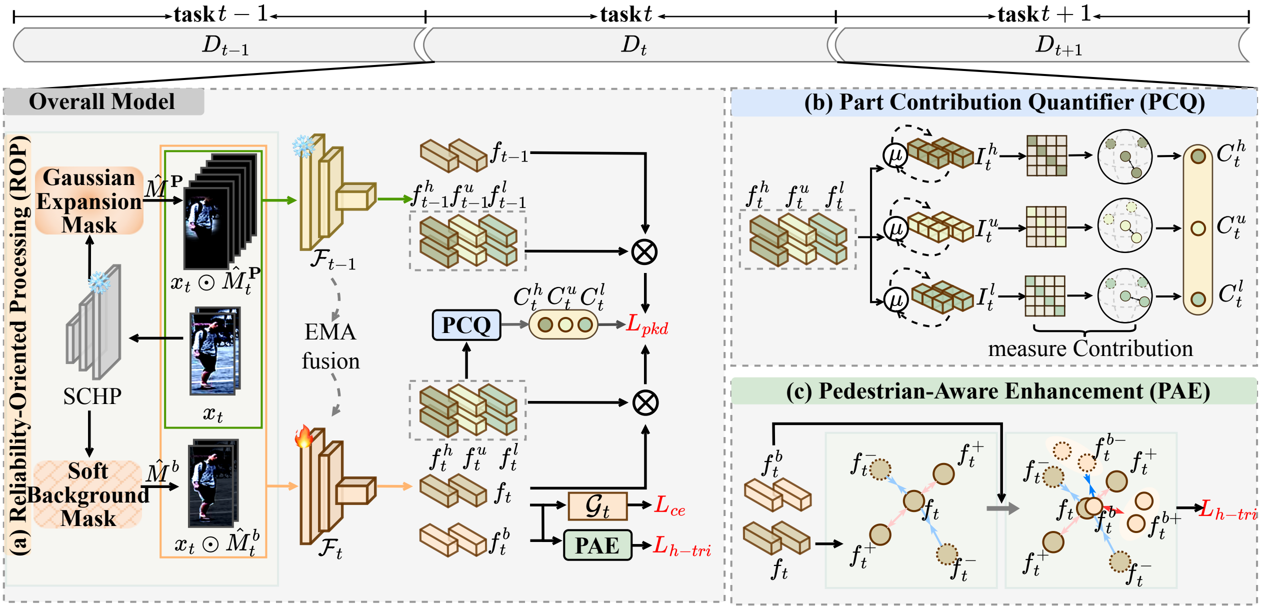


Fig. 3. The overall framework of our proposed PKTA. First, images x_t are input to **ROP** (a) to refine pedestrian parts, then fed into the new model $\mathcal{F}_t$ and old model $\mathcal{F}_{t-1}$ to extract their respective features. Next, **PCQ** (b) granularly explores parts and obtains their contributions $C_t^{\mathbf{P}}$ to guide the selective part-level knowledge transfer. Finally, **PAE** (c) suppresses background noise and encourages the model to focus on acquiring pedestrian regional information.

A. *Problem Definition*

In the Non-Exemplar LReID setting, different domain datasets form a task sequence simulating non-stationary data $\mathcal{D} = \{D_t\}_{t=1}^{T}$. Here, $D_t = \{(x_{t,i}, y_{t,i})\}_{i=1}^{N_t}$ contains N_t instances $x_{t,i}$ and labels $y_{t,i}$. For Task t, the model $\mathcal{F}_t$ cannot access instances from previous t-1 tasks. In contrast to [28] and [17], which rely on revisiting prototype representations for knowledge retention, our method avoids any historical feature revisiting, thus eliminating inherent storage constraints. The model $\mathcal{F}_t$ is then evaluated separately on seen data $\mathcal{D}^{test} = \{D_i^{test}\}_{i=1}^{t}$ and unseen data $\mathcal{D}^{un} = \{D_i^{un}\}_{i=1}^{U}$, where U is the number of unseen datasets.

B. *Reliability-Oriented Processing (ROP)*

For an input image $x_{t,i} \in \mathbb{R}^{3\times H\times W}$, where H and W are image height and width, we first use the pretrained semantic segmentation model SCHP [10] to parse the image into fine-grained pedestrian part masks. Specifically, two categories of masks are generated. One is the "body" $M_i^b \in \mathbb{R}^{1\times H\times W}$ covers the entire pedestrian region, and the other includes three local masks "head", "upper" and "lower", collectively $M_i^{\mathbf{P}} \in \mathbb{R}^{1\times H\times W}$, $\mathbf{P} \in \{h, u, l\}$. Formally, both M_i^b and $M_i^{\mathbf{P}}$ binary map $\Omega \to \{0, 1\}$, where 0 denotes that the corresponding pixel belongs to the background region, while 1 denotes a foreground pixel that falls within the target pedestrian part region. $\Omega \subseteq \mathbb{Z}^2$ is the set of all image pixel coordinates. Despite the effectiveness of the SCHP model in rough part segmentation, the initially obtained part semantics suffer from non-negligible deficiencies such as blurred part boundaries and inaccurate pixel-level localization, as visually illustrated in Figure 4. To address this limitation and enhance the reliability of part-level feature exploration in subsequent steps, we introduce a prior pixel distribution tailored to each pedestrian part. This distribution is derived from statistical characteristics of pedestrian anatomy and spatial layout, which serves to refine the initial coarse masks and yield more precise, semantically consistent part segmentation results.

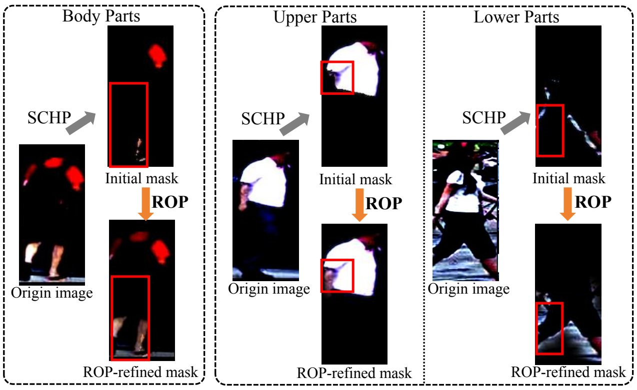
Soft Background Mask: For a given body mask M_i^b generated by the SCHP model, ambiguous pedestrian boundaries or model prediction bias may cause the initial mask to fail to fully cover the complete pedestrian body region. To rectify the incomplete segmentation, we do not simply discard the pixels outside the initially covered area. Instead, we implement a targeted pixel retention based on the spatial adjacency of foreground pixels and the prior anatomical structure of human bodies. Specifically, we first identify the edge pixels of the initial mask M_i^b and then expand the mask region by retaining the adjacent non-background pixels that are spatially consistent with the overall pedestrian silhouette. Through this refinement process, the optimized body mask $\hat{M}_i^b$ is obtained, which strictly maintains the structural integrity of the pedestrian body region:

$$\hat{M}_i^b(\omega) = (1 - M_i^b(\omega)) \times (W_i^b M_i^b(\omega)) + M_i^b(\omega), \quad (1)$$

$$W_i^b = 1 - exp(-\gamma|\frac{\sum_{\omega\in\Omega}\mathbb{I}(M_i^b(\omega)=1)}{H\times W} - \epsilon|), \quad (2)$$

where $\omega \in \Omega$ is a pixel coordinate. $\epsilon = 0.6$ is the approximate proportion of pedestrian body derived from prior statistics[1], $\gamma = 10$ regulates the retention extent of

[1]Our preliminary statistics show that the area ratio of the pedestrian body is approximately 0.6 in most images from relevant datasets.



(a) Soft Background Mask (b) Gaussian Expansion Mask

Fig. 4. Initial SCHP-output masks vs. ROP-refined masks. SCHP-output masks perform coarse-grained segmentation and easily miss pedestrian information, while ROP-refined masks enable fine-grained segmentation and fully retain pedestrian feature details.

non-predicted pixels. This refinement is critical for avoiding information loss of key body parts and laying a solid foundation for the subsequent enhancement of pedestrian region knowledge acquisition.

Gaussian Expansion Mask: Regarding the other part masks $M_i^{\mathbf{P}}$, we apply Gaussian expansion to these masks to achieve smooth boundary gradation, ensuring the refined mask $\hat{M}_i^{\mathbf{P}}$ better aligns with the true distribution of pedestrian parts. First, we calculate the minimum squared Euclidean distance $Ed(\omega)$ between the target pixel ω and the nearest foreground pixel j. This metric effectively quantifies the spatial correlation between each pixel and the core region of the target pedestrian part:

$$Ed(\omega) = \min_{j \in \Omega, M_i^{\mathbf{P}}(j)=1} \|\omega - j\|^2. \tag{3}$$

Then, we assign a gradient weight to these background pixels, where the weight decreases exponentially with the distance from the foreground core, realizing the smooth boundary transition:

$$\hat{M}_i^{\mathbf{P}}(\omega) = (1 - M_i^{\mathbf{P}}(\omega))e^{-\frac{1}{2\sigma}Ed(\omega)} + M_i^{\mathbf{P}}(\omega), \tag{4}$$

where $\sigma = 10.0$ is the smoothing parameter. This enhances the consistency between the refined mask and the true anatomical structure of pedestrian parts for fine-grained part-level knowledge transfer in subsequent steps.

C. Part Contribution Quantifier (PCQ)

In this section, since traditional knowledge distillation impairs model stability via redundant old knowledge and compromises plasticity by occupying new knowledge learning capacity, we achieve selective part-level knowledge transfer by measuring the contributions of pedestrian parts. During an iteration of task t with batch size N_b, given data $\{(x_{t,i}, y_{t,i})\}_{i=1}^{N_b}$ sampled from D_t and their refined masks $\hat{M}^{\mathbf{P}}$, the new model $\mathcal{F}_t$ and the old model $\mathcal{F}_{t-1}$ extract the global features $f_t = \mathcal{F}_t(x_t)$ and $f_{t-1} = \mathcal{F}_{t-1}(x_t)$, and the part-guided features $f_t^{\mathbf{P}} = \mathcal{F}_t(\hat{M}^{\mathbf{P}} \odot x_t)$ and $f_{t-1}^{\mathbf{P}} = \mathcal{F}_{t-1}(\hat{M}^{\mathbf{P}} \odot x_t)$, respectively.

First, we model the part-guided features $I_t^{\mathbf{P}} \in \mathbb{R}^{N_t^I \times d}$ for each identity, where N_t^I is the identity number in task t and d is the feature dimension. We perform dynamic cumulative updates to capture more comprehensive representations of each identity over training iterations:

$$I_{t,i}^{\mathbf{P}}[B] = \mu I_{t,i}^{\mathbf{P}}[B-1] + (1-\mu)\frac{1}{K_{B,i}}\sum_{k=1}^{K_{B,i}} f_{t,k}^{\mathbf{P}}, \tag{5}$$

where $K_{B,i}$ means the number of instances of the i-th identity in the B-th iteration and $\mu = 0.9$ is momentum coefficient. Then, we quantify the contribution $C_t^{\mathbf{P}}$ of each part via the separability of part-guided features across identities, where higher separability indicates greater contribution to distinguishing identities:

$$C_t^{\mathbf{P}} = \frac{1}{N_b}\sum_{j=1}^{N_b}[-log(\frac{e^{(cos(f_{t,j}^{\mathbf{P}}, I_{t,y_{t,j}}^{\mathbf{P}})/\tau)}}{\sum_{i=1}^{N_t^I} e^{(cos(f_{t,j}^{\mathbf{P}}, I_{t,i}^{\mathbf{P}})/\tau)}})], \tag{6}$$

where $\tau = 0.1$ is temperature parameter and $cos(\cdot)$ is cosine similarity. And the part-level contribution $C_t^{\mathbf{P}}$ guides selective knowledge transfer, where higher-contribution parts are imposed with weaker transfer constraints, allowing the model to flexibly learn new knowledge, while lower-contribution parts tend to transfer general old knowledge from the previous model:

$$L_{pkd} = \sum_{\mathbf{P} \in \{h,u,l\}} \mathbf{W}^{\mathbf{P}} \mathcal{L}_{KL}(\rho(\frac{f_{t-1}^{\mathbf{P}} f_{t-1}^{\top}}{\lambda}) \| \rho(\frac{f_t^{\mathbf{P}} f_t^{\top}}{\lambda})), \tag{7}$$

where $\mathcal{L}_{KL}$ is the Kullback Leibler divergence, $\rho(\cdot)$ is the softmax function applied row-wise, $\lambda = 0.1$ is a temperature parameter, and $\mathbf{W}^{\mathbf{P}}$ denotes the normalized contribution of each part, which serves to modulate the selective knowledge transfer:

$$\mathbf{W}^{\mathbf{P}} = \frac{1/C_t^{\mathbf{P}}}{\sum\limits_{j \in \{h,u,l\}} 1/C_t^j}. \tag{8}$$

D. Pedestrian-Aware Enhancement (PAE)

To enhance pedestrian region knowledge acquisition and mitigate dynamic background noise during task transitions, we drive the clustering of global features from the same identity and leverage body features $f_t^b = \mathcal{F}_t(\hat{M}^b \odot x_t)$ to provide auxiliary guidance for the model to focus more on body regions, thereby mitigating the noise induced by background variations during task transitions. Specifically, PAE first enforces the clustering of global features from samples of the same pedestrian identity to enhance the model's discriminative ability for holistic pedestrian representation. Concurrently, we leverage the body-region features f_t^b to provide explicit auxiliary guidance. This guidance drives the model to prioritize the learning of discriminative information from the complete pedestrian body region, minimizing the adverse impact of irrelevant non-pedestrian information on feature representation:

$$\begin{aligned} L_{h-tri} &= log(1 + exp(\|f_t - f_t^{+}\|_2^2 - \|f_t - f_t^{-}\|_2^2)) \\ &+ log(1 + exp(\|f_t^b - f_t^{b+}\|_2^2 - \|f_t^b - f_t^{b-}\|_2^2)) \end{aligned}, \tag{9}$$

TABLE I
QUANTITATIVE COMPARISON RESULTS ON TRAINING ORDER-1: MARKET-1501 → CUHK-SYSU → DUKEMTMC → MSMT17 → CUHK03. **BOLD** VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES DENOTE THE SECOND-BEST PERFORMANCE.

	Method	Non-Exemplar	Market-1501		CUHK-SYSU		DukeMTMC		MSMT17		CUHK03		**Seen-Avg**		**UnSeen-Avg**	
			mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
CIL	LwF [15]		56.3	77.1	72.9	75.1	29.6	46.5	6.0	16.6	36.1	37.5	40.2	50.6	47.2	42.6
CIL	SPD [23]		35.6	61.2	61.7	64.0	27.5	47.1	5.2	15.5	42.2	44.3	34.4	46.4	40.4	36.6
CIL	PRAKA [21]	✓	37.4	61.3	69.3	71.8	35.4	55.0	10.7	27.2	**54.0**	**55.6**	41.3	54.2	47.7	41.6
CIL	FCS [11]	✓	58.3	78.5	75.1	76.2	42.6	59.8	10.2	24.3	35.3	34.9	44.3	54.7	52.1	44.2
LReID	CRL [32]		58.0	78.2	72.5	75.1	28.3	45.2	6.0	15.8	37.4	39.8	40.5	50.8	47.8	43.5
LReID	AKA [18]		51.2	72.0	47.5	45.1	18.7	33.1	16.4	37.6	27.7	27.6	32.3	43.1	44.3	40.4
LReID	PatchKD [22]		**68.5**	**85.7**	75.6	78.6	33.8	50.4	6.5	17.0	34.1	36.8	43.7	53.7	49.1	45.4
LReID	MEGE [19]		39.0	61.6	73.3	76.6	16.9	30.3	4.6	13.4	36.4	37.1	34.0	43.8	47.7	44.0
LReID	CKP [27]	✓	53.8	76.0	81.2	83.0	49.7	67.0	18.4	40.8	44.1	45.8	49.4	62.5	58.0	51.0
LReID	LSTKC [29]	✓	54.7	76.0	81.1	83.4	49.4	66.2	20.0	43.2	44.7	46.5	50.0	63.1	57.0	49.9
LReID	DKP [28]	✓	60.3	80.6	83.6	**85.4**	51.6	68.4	19.7	41.8	43.6	44.2	51.8	64.1	59.2	51.6
LReID	PKTA	✓	63.2	81.9	**83.7**	83.9	**53.5**	**69.4**	**20.3**	**43.5**	48.3	50.8	**53.8**	**65.9**	**59.8**	**53.0**

TABLE II
QUANTITATIVE COMPARISON RESULTS ON TRAINING ORDER-2: DUKEMTMC → MSMT17 → MARKET-1501 → CUHK-SYSU → CUHK03. **BOLD** VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES DENOTE THE SECOND-BEST PERFORMANCE.

	Method	Non-Exemplar	DukeMTMC		MSMT17		Market-1501		CUHK-SYSU		CUHK03		**Seen-Avg**		**UnSeen-Avg**	
			mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
CIL	LwF [15]		42.7	61.7	5.1	14.3	34.4	58.6	69.9	73.0	34.1	34.1	37.2	48.4	44.0	40.1
CIL	SPD [23]		28.5	48.5	3.7	11.5	32.3	57.4	62.1	65.0	43.0	45.2	33.9	45.5	39.8	36.3
CIL	PRAKA [21]	✓	31.2	48.7	6.6	19.1	47.8	69.8	70.4	73.0	**54.9**	**56.6**	42.2	53.4	48.4	41.1
CIL	FCS [11]	✓	53.6	70.0	9.5	23.5	48.7	69.9	76.2	78.2	37.1	38.4	45.0	56.0	52.7	45.1
LReID	CRL [32]		43.5	63.1	4.8	13.7	35.0	59.8	70.0	72.8	34.5	36.8	37.6	49.2	45.3	41.4
LReID	AKA [18]		32.5	49.7	-	-	-	-	-	-	-	-	-	-	40.8	37.2
LReID	PatchKD [22]		**58.3**	**74.1**	6.4	17.4	43.2	67.4	74.5	76.9	33.7	34.8	43.2	54.1	48.6	44.1
LReID	MEGE [19]		21.6	35.5	3.0	9.3	25.0	49.8	69.9	73.1	34.7	35.1	30.8	40.6	44.3	41.1
LReID	CKP [27]	✓	49.4	67.0	14.5	33.8	56.0	77.6	83.2	84.9	45.3	47.1	49.7	62.1	57.2	50.0
LReID	LSTKC [29]	✓	49.9	67.6	14.6	34.0	55.1	76.7	82.3	83.8	46.3	48.1	49.6	62.1	57.6	49.6
LReID	DKP [28]	✓	53.4	70.5	14.5	33.3	60.6	81.0	83.0	84.9	45.0	46.1	51.3	63.2	59.0	51.6
LReID	PKTA	✓	52.7	70.2	**14.7**	**34.4**	**64.1**	**82.8**	**84.1**	**85.8**	48.1	50.6	**52.7**	**64.8**	**59.4**	**51.6**

where $\langle f_t^b, f_t^{b+}, f_t^{b-}\rangle$ and $\langle f_t, f_t^+, f_t^-\rangle$ are triplets for body and global features, respectively.

E. Training and Inference

The overall loss L comprises L_{ce}, L_{h-tri} and L_{pkd}. Specifically, L_{ce} and L_{h-tri} are primarily designed to facilitate the learning of discriminative representations for novel pedestrian identities in the current incremental task, while L_{pkd} is dedicated to enabling effective knowledge transfer for previously learned pedestrian identities to mitigate catastrophic forgetting:

$$L = L_{ce} + \alpha L_{h-tri} + L_{pkd}, \tag{10}$$

where $\alpha = 0.5$ is a balancing parameter for loss weights (Ablation studies are presented in Section IV-D). Among them,

$$L_{ce} = -y_t log(\rho(\mathcal{G}_t(f_t))), \tag{11}$$

is the cross-entropy loss, where $\mathcal{G}_t$ as linear projection mapping features to logits.

To further consolidate new and old knowledge, following common practices [29], we employ Exponential Moving Average (EMA) fusion as post-processing:

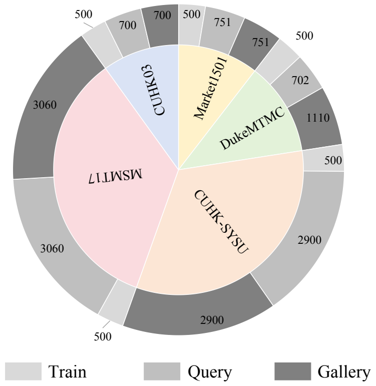
$$\mathcal{F}_t \leftarrow (\mathcal{F}_t + \mathcal{F}_{t-1})/2. \tag{12}$$

During the inference phase, we exclusively employ the final model $\mathcal{F}_T$ to extract features from pedestrian instances and perform retrieval.

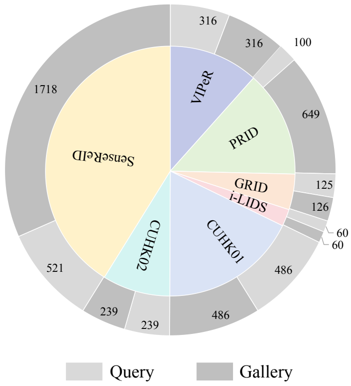
IV. EXPERIMENTS

A. Setup

Datasets: All experiments are conducted in adherence to the LReID benchmark [18]. The seen datasets include Market-1501 [34], CUHK-SYSU [26], DukeMTMC [20], MSMT17 [24], and CUHK03 [14], which are employed



(a) Seen domains



(b) Unseen domains

Fig. 5. Statistics of pedestrian ID numbers across different datasets on seen and unseen domains.

for both training and testing. To assess the generalization capability of the methods, the unseen datasets, including CUHK01 [13], CUHK02 [12], GRID [16], SenseReID [33], VIPeR [6], i-LIDS [3], and PRID [7], are exclusively utilized for testing. The detailed pedestrian count statistics across the datasets are presented in Figure 5. To comprehensively evaluate the methods, we consider two training orders, specified as follows:

- **Training Order-1**: Market-1501 → CUHK-SYSU → DukeMTMC → MSMT17 → CUHK03;
- **Training Order-2**: DukeMTMC → MSMT17 → Market-1501 → CUHK-SYSU → CUHK03.

Evaluation Metrics: Following the general experimental settings [18], we evaluate performance using mean Average Precision (mAP) and Rank@1 accuracy (R@1) on each dataset. Additionally, we report the average mAP and average R@1 across both seen and unseen datasets to compare overall performance and assess generalization ability to unseen identities.

B. Implementation Details

We use ResNet50 as the backbone of our method. The input image size is 256×128, with data augmentation including horizontal flipping, random erasing, and random cropping. For each training order, we train the model for 80 epochs with a batch size of 128, where each batch comprises 4 instances per identity. SGD is employed as the optimizer, configured with a learning rate of 0.008 and a weight decay of 1e-4. For the SCHP model, we adopt its pretrained variant with seven pedestrian part partitions, where we merge three parts (torso, upper arms, and lower arms) into a unified upper body region, while grouping thighs and calves into a single lower body region. All experiments are implemented in PyTorch and executed on an NVIDIA RTX 4090 GPU.

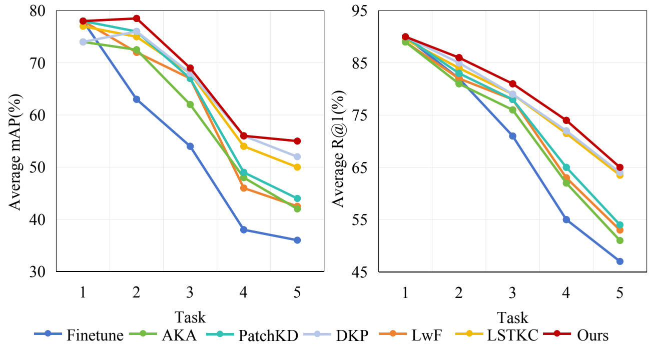


Fig. 6. Performance curves on seen domains under training order-1. The model's performance across all seen domains is evaluated after each task training.

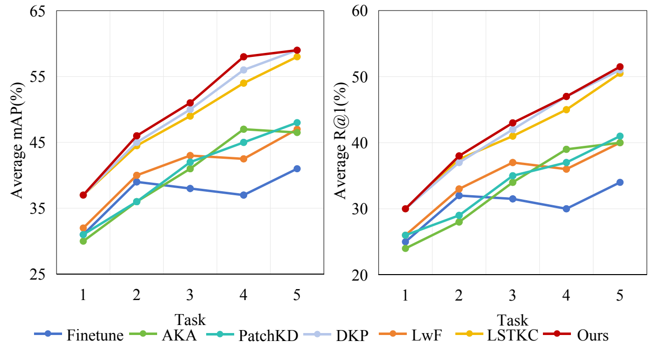


Fig. 7. Performance curves on unseen domains under training order-1. The model's performance across all unseen domains is evaluated after each task training.

C. Comparison with the State-Of-The-Art

We compare our PKTA with state-of-the-art LReID approaches: CRL [32], AKA [18], PatchKD [22], MEGE [19], CKP [27], LSTKC [29], and DKP [28]. To provide a broader context, we also include CIL methods: LwF [15], SPD [23], PRAKA [21], and FCS [11].

As shown in Tables I and II, our PKTA performs competitively across two training orders in both seen and unseen domains, validating its effectiveness. Specifically, under training order-1, our proposed method improves the average mAP and the average R@1 relative to the top-performing methods by **2.0%** and **1.8%** on seen scenarios. On unseen scenarios, our method achieves improvements of **0.6%** in average mAP and **1.4%** in average R@1. For training order-

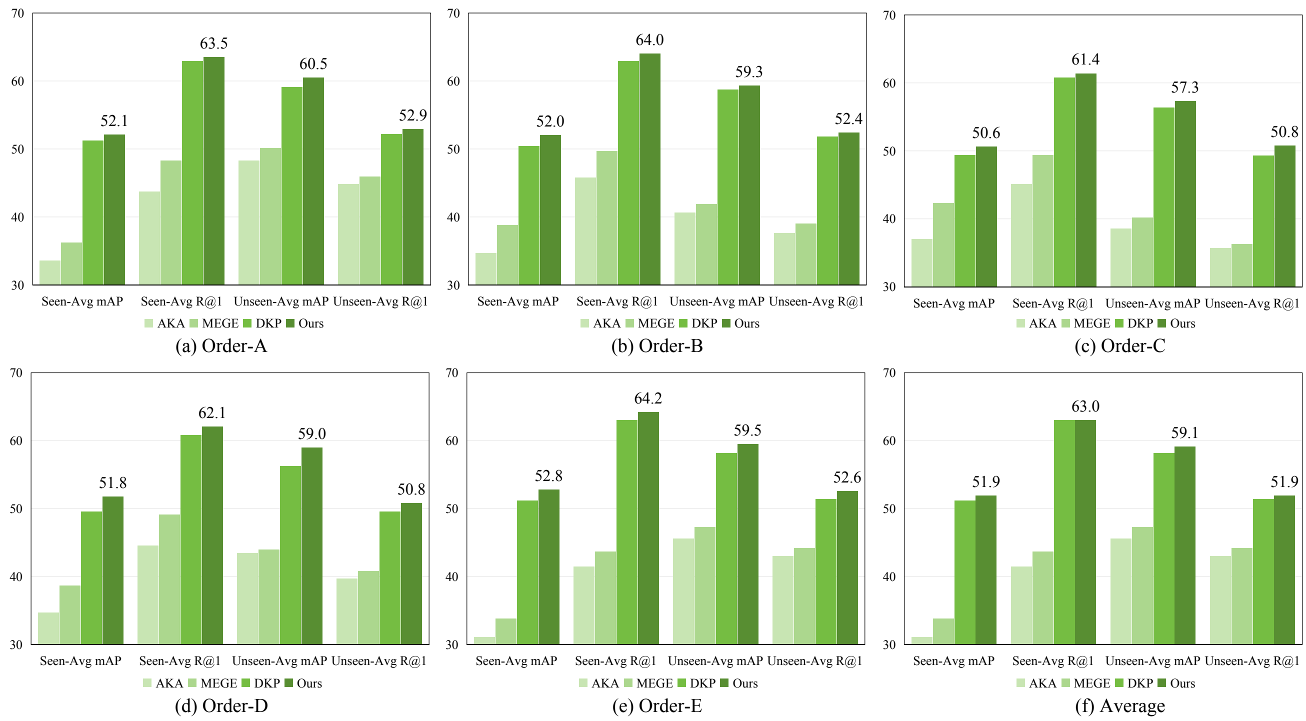


Fig. 8. Performance comparison on different training orders. (a) Order-A: MSMT17 → CUHK-SYSU → DukeMTMC → Market-1501 → CUHK03. (b) Order-B: DukeMTMC → Market-1501 → CUHK03 → MSMT17 → CUHK-SYSU. (c) Order-C: CUHK-SYSU → DukeMTMC → CUHK03 → MSMT1 → Market-1501. (d) Order-D: CUHK03 → MSMT17 → DukeMTMC → Market-1501 → CUHK-SYSU. (e) Order-E: Market-1501 → MSMT17 → DukeMTMC → CUHK-SYSU → CUHK03. (f) The average performance of the above five orders.

2, compared with SOTA DKP [28], our approach attains gains of **1.4%** in average mAP and **1.6%** in average R@1 on seen scenarios. It also yields a **0.4%** gain in average mAP while achieving parity with DKP in average R@1 on unseen scenarios. Notably, PatchKD [22] achieves optimal performance on the first task, which benefits from its strict constraint of knowledge distillation. However, this constraint hinders new task learning, decreasing subsequent task performance. Moreover, while PRAKA [21] performs better on the last task, our method outperforms it on previous tasks and achieves higher average performance across both domains by focusing on pedestrian parts information.

Performance Curves: Figure 6 and Figure 7 illustrate the performance curves on seen and unseen domains, respectively, as incremental training progresses. On the seen domains, our method achieves favorable initial accuracy on the first task. This performance gain is attributed to the PAE module, which leverages body-region features to constrain the model to prioritize the extraction of pedestrian information while minimizing interference from dynamic background variations. This targeted guidance effectively contributes to the initial performance boost. Furthermore, our method sustains superior performance across subsequent tasks, which demonstrates its effectiveness in mitigating catastrophic forgetting and its robust capability to consolidate both previously learned and newly acquired pedestrian identity knowledge. Then, on the unseen domains, our method consistently achieves the best performance as more tasks are learned, validating its strong generalization capability to previously unseen pedestrian identities.

D. Ablation Study

Influence of Different Components: We conduct the ablation experiments to evaluate our method's component performance in Table III. *Baseline* excludes ROP, PAE, and PCQ. ROP refines pedestrian part extraction to lay the foundation for PAE and PCQ to enhance pedestrian regional knowledge acquisition and facilitate selective part-level knowledge transfer, respectively. Incorporating these components gradually improves performance. Our observations are as follows: (1) ROP&PAE suppresses noise interference caused by background variations during task incrementation, enabling the model to focus on pedestrian regions. Compared with the baseline, it improves the average mAP/R@1 by **2.0%/3.1%** on seen scenarios and by **4.2%/2.6%** on unseen scenarios, respectively. (2) ROP&PCQ performs differential knowledge transfer across different pedestrian parts, thereby achieving fine-grained and thorough exploration of pedestrian information. It yields an average improvement of **8.9%** in mAP and **9.5%** in R@1 over the baseline, on both seen and unseen scenarios. (3) Our PKTA, which includes ROP, PAE, and PCQ, exerts a synergistic effect among its components and thus achieves the best performance.

Evaluation of Different Training Orders: Beyond the two widely adopted training orders in the existing evaluation, we further conduct a comprehensive comparison of

TABLE III

ABLATION STUDY OF THE RELIABILITY-ORIENTED PROCESSING (ROP), PEDESTRIAN-AWARE ENHANCEMENT (PAE) AND PART CONTRIBUTION QUANTIFIER (PCQ) MODULES IN TRAINING ORDER-1.

	Baseline	ROP	PAE	PCQ	Seen-Avg		Unseen-Avg	
					mAP	R@1	mAP	R@1
Order-1	✓				44.7	54.8	49.0	42.5
	✓	✓	✓		46.7	57.9	53.2	45.1
	✓	✓		✓	52.5	64.6	59.0	51.7
	✓	✓	✓	✓	53.8	65.9	59.8	53.0

TABLE IV

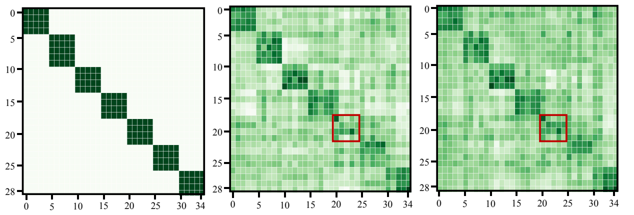
ABLATION STUDY ON BALANCING PARAMETERS α FOR THE OVERALL LOSS ACROSS TRAINING ORDER-1 AND TRAINING ORDER-2.

	loss weigtht	Seen-Avg		Unseen-Avg	
		mAP	R@1	mAP	R@1
Order-1	α=0.1	53.1	65.6	59.2	52.4
	α=0.5	53.5	65.7	59.8	53.0
	α=1.0	53.2	65.2	58.9	51.7
Order-2	α=0.1	51.9	63.7	58.3	50.7
	α=0.5	52.7	64.8	59.4	51.6
	α=1.0	52.1	64.2	58.8	50.9

different methods under multiple training orders to more thoroughly validate the effectiveness of our approach. As shown in Figure 8, our method consistently achieves the optimal performance across all evaluated training orders. In terms of average performance, our method surpasses DKP by **1.5%** and **0.9%** in mAP and R@1 on seen domains, respectively, and by **1.4%** and **1.0%** in mAP and R@1 on unseen domains. These results indicate that our method is robust to different training-order sequences and adapts well across diverse scenarios.

Impact of the α: We analyze the impact of the hyperparameter for L_{pkd} separately under training order-1 and training order-2. As illustrated in Table IV, variations in the hyperparameter α exert a noticeable impact on the experimental results. In particular, the model achieves its optimal performance across all evaluation metrics when $\alpha = 0.5$. Mechanistically, L_{pkd} is designed to facilitate knowledge transfer for preserving previously learned pedestrian identities. When the weight of L_{pkd} is excessively large, the model may suffer from over-constraint, which in turn leads to the neglect of learning new pedestrian identities in the current incremental task. Conversely, an overly small weight of L_{pkd} fails to effectively retain the knowledge of historical identities, thereby impairing the model's anti-forgetting capability.

Visualization Studies: We visualize the relation matrix on *DukeMTMC* test images to intuitively validate the effectiveness of our proposed PKTA in Figure 9. Specifically, the *Query* and *Gallery* sets are sampled from seven pedestrian identities, containing 29 and 35 instances, respectively. (a) shows the ground truth derived from annotated labels. In contrast, conventional knowledge distillation (b) yields ambiguous label predictions within the red boxes, whereas our PKTA (c) produces clearer predictions in these regions. This visualization demonstrates that our method better maintains and transfers pedestrian identity information across long-term task sequences, thereby more effectively adapting to the LReID.



(a) Ground Truth (b) Distillation (c) Our PKTA

Fig. 9. Relation matrix of test images from DukeMTMC on the final model trained under training order-1.

V. CONCLUSION

This paper proposes the Part-oriented Knowledge Transfer and Acquisition (PKTA) strategy for Non-Exemplar Lifelong Person Re-Identification. To balance the model's stability and plasticity, we first design a Reliability-Oriented Processing (ROP) module. This module refines the coarse-grained pedestrian part masks generated by a pretrained SCHP model. Subsequently, a Part Contribution Quantifier (PCQ) module is introduced to enable selective part-level knowledge transfer, preserving general old knowledge accumulated from previous tasks while avoiding interference with critical new knowledge learning. Moreover, the Pedestrian-Aware Enhancement (PAE) module guides the model to focus on pedestrian regions, reducing the background noise during task transition and enhancing high quality pedestrian knowledge acquisition. Extensive experiments across different training orders validate PKTA's effectiveness.

For future work, we will investigate how the number of pedestrian part partitions affects model performance. Specifically, leveraging the SCHP model's fine-grained human parsing capability (*e.g.*, 7-class semantic segmentation), we will examine whether coarse-to-fine part partitioning for knowledge transfer helps the model capture richer pedestrian information, or instead degrades performance by overemphasizing part-level knowledge at the expense of holistic identity representations.

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